

Manufacturing Technology 2022

Program: Diploma in Mechanical Engineering / Mechatronics. Semester 2. Revision 2021. Credits 3.
Engineering Science. Periods: 3 per week, 45 per semester.

This downloadable handbook covers foundry, casting, metal working, powder metallurgy, welding, brazing, soldering, forging and press working. Use this with the full lesson HTML for diagrams and interactive navigation.

Overview and roadmap

Objectives: understand the importance of manufacturing process in day-to-day life and study basic manufacturing process; familiarize conventional manufacturing processes like casting, welding and metal forming.

Prerequisite: Engineering Workshop Practice, basic manufacturing process, foundry and welding. CO1 foundry relevance 11 h; CO2 casting and working 10 h; CO3 metal joining 12 h; CO4 forging and press working 10 h.

CO-PO mapping: CO1, CO2 and CO4 moderately mapped to PO1. CO3 moderately mapped to PO1 and PO4. Official model question pattern is not specified in the supplied syllabus/source.

Module 1 - Foundry basics

Manufacturing converts raw material into useful product by changing shape, size, surface, property or joining condition. Mechanical properties: strength, hardness, brittleness, toughness, malleability, ductility, stiffness, fatigue and resilience.

Patterns: single piece, split, match plate, gated, loose piece and sweep. Pattern materials: wood, metal, plastic and plaster. Allowances: shrinkage, draft, machining, distortion/camber and rapping.

Moulding sand properties: porosity, plasticity, adhesiveness, cohesiveness and refractoriness. Sand types: green, dry, parting, loam, facing and core sand. Moulding: bench, floor, pit and sweep. Draw gating system with pouring basin, sprue, runner, gate, cavity, riser and chiller.

Module 2 - Casting and working

Casting types: ferrous, non-ferrous and special. Processes: centrifugal casting, die casting, gravity die casting, pressure die casting, hot chamber/goose-neck/direct injection, cold chamber, slush casting and vacuum casting.

Defects: blow holes, shrinkage cavity, cold shut, misrun, hot tear, sand inclusion and mismatch. Remedies: correct sand moisture, venting, pouring temperature, gating, riser and mould alignment.

Cold working below recrystallisation temperature gives good finish and strain hardening but needs high force. Hot working above recrystallisation temperature needs lower force and refines grains but causes scale. Powder metallurgy stages: powder production, blending, compaction, sintering and finishing.

Module 3 - Joining processes

Welding classification: fusion welding and pressure welding. Gas welding uses neutral, carburizing and oxidizing oxy-acetylene flames. Arc welding uses AC transformer or DC generator, electrode holder, cables, earth clamp and protective equipment.

Processes: SMAW, submerged arc, TIG, MIG, resistance spot, seam, projection and thermit welding. Electrode coating stabilizes arc, shields weld, forms slag, deoxidizes and can add alloying elements.

Defects: porosity, slag inclusion, undercut, crack, incomplete penetration, distortion and spatter. Brazing uses filler above 450 deg C; soldering uses filler below 450 deg C. Base metal is not melted in either process.

Module 4 - Forging and press working

Forging shapes metal by compressive force. Flat die/open die forging uses simple dies; closed die forging uses shaped cavities. Tools: anvil, swage block, hammers, tongs, chisels, swages, fullers, flatter, set hammer, punch and drift.

Press working operations: cutting, bending, drawing, punching, blanking, notching and lancing. Die-set parts: punch, die shoe, guide pin, bolster plate, stripper, stock guide, feed stock and pilot.

Punching makes the useful hole and slug is scrap. Blanking makes the useful blank and remaining strip is scrap. Clearance affects burr, cutting force, edge quality and tool life. No numerical problems are prescribed.

Rule bank and solved examples

Rules: pattern is not the mould; riser feeds shrinkage; chiller increases local cooling; cold working is below recrystallisation temperature; hot working is above it; punching gives a hole; blanking gives a blank.

Solved example 1: A pump housing with internal cavity is best made by sand casting with core, gating and riser. Reason: complex cavity and economical production compared with machining from solid.

Solved example 2: A sheet-metal washer needs blanking for outer profile and punching for centre hole. Solved example 3: Porosity in weld bead can be due to moisture, dirty surface or poor shielding; remedy by cleaning, drying electrodes and correct arc length.

Practice, model paper and answers

Expected questions: explain pattern allowances; draw gating system; compare hot and cold working; explain powder metallurgy; classify welding; draw arc welding; explain electrode coating; explain forging tools; compare punching and blanking.

Fresh model paper: define manufacturing process; list mechanical properties; state riser function; explain pattern materials and allowances; explain centrifugal and die casting; compare hot/cold working; classify welding; explain gas welding flames; draw arc welding; explain resistance welding; explain forging and press operations.

Answer guide: include definition, principle, labelled sketch, sequence, advantages/limitations, applications and defects/safety. Use keywords: shrinkage allowance, draft, porosity, refractoriness, recrystallisation temperature, fusion welding, pressure welding, electrode coating, die clearance.

References

T1 Manufacturing process - Begeman - 5th Edition - McGraw Hill, New Delhi 1981. R1 Workshop Technology - W A J Chapman - Volume I, II and III. R2 Production Technology - HMT - Edn. 18 - Tata McGraw Hill - 2001.

R3 Production Technology - P. C. Sharma - Edn. X - S. Chand - 2006. R4 Production Technology, Edn. XII, Khanna Publishers - 2006. Online resources: <https://nptel.ac.in/courses/112107144/> and <https://nptel.ac.in/courses/112104195/>